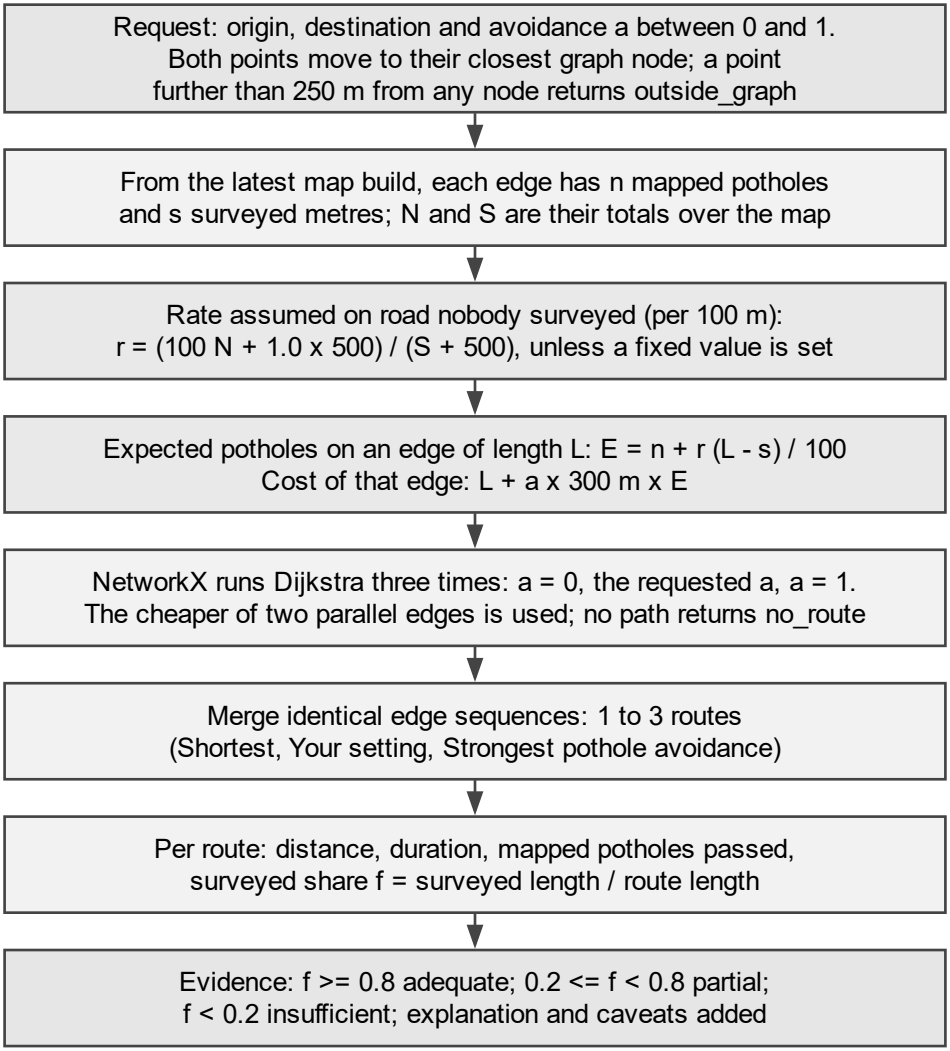
Request: origin, destination and avoidance a between 0 and 1.
Both points move to their closest graph node; a point
further than 250 m from any node returns outside_graph
From the latest map build, each edge has n mapped potholes
and s surveyed metres; N and S are their totals over the map
Rate assumed on road nobody surveyed (per 100 m):
r = (100 N + 1.0 x 500) / (S + 500), unless a fixed value is set
Expected potholes on an edge of length L: E = n + r (L - s) / 100
Cost of that edge: L + a x 300 m x E
NetworkX runs Dijkstra three times: a = 0, the requested a, a = 1.
The cheaper of two parallel edges is used; no path returns no_route
Merge identical edge sequences: 1 to 3 routes
(Shortest, Your setting, Strongest pothole avoidance)
Per route: distance, duration, mapped potholes passed,
surveyed share f = surveyed length / route length
Evidence: f >= 0.8 adequate; 0.2 <= f < 0.8 partial;
f < 0.2 insufficient; explanation and caveats added